

netflow: formulation, closure models, and verification of a generic scalar conservation-network solver with a quantitative validation protocol against the VERA core-physics benchmark

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Abstract

This is a methods companion to the introductory note on **netflow**. It specifies what the introductory note asserted: the residual equations that are assembled, how boundary conditions are imposed, the closure models used in each physics, and the verification evidence. **netflow** solves any problem reducible to *conservation of a scalar flux at the nodes of a network*. A domain-agnostic core assembles the nodal residual $F_i = s_i + \sum \text{flux}_{\text{in}} - \sum \text{flux}_{\text{out}}$ and solves $F(\mathbf{x}) = 0$ by damped Newton over a sparse analytic Jacobian, with a Picard fallback. Larger problem structures—transient, eigenvalue, coupled multiphysics—are expressed as an *outer loop* of driven core solves. We give the governing residuals for the thermal, hydraulic, and neutronic plugins; the boundary-condition idioms (Dirichlet, Neumann, advective outflow, $\varphi=0$ ground, eigenvalue closure); and the closure correlations (temperature-dependent solid conductivity, fuel-pellet and gap models, single-phase convection, gray-body radiation, and a calibrated cross-flow network). Verification is exact against analytic closed forms, with second-order spatial convergence of the neutron eigenvalue and a 107-test automated suite. Section 7 replaces the note’s qualitative “matches the pattern” claim with a quantitative *code-to-code* comparison against published VERA solutions—the benchmark’s Problems 6 and 7 have no reference solution—tabulating each quantity at its published definition (volume-average, not centerline, fuel temperature; figure-derived coolant spread) with input and calibrated quantities marked. Peak volume-average fuel temperature agrees with the published MC21/CTF solution to within 6%; the calibrated subchannel spread brackets the published code spread. The tool’s modeling boundary is stated explicitly: it carries no momentum equation and no two-phase model, and is positioned as a transparent pre-design scoping tool, not a replacement for a subchannel code such as CTF/COBRA-TF.

1 Scope of this document

The companion note motivated **netflow** and showed that one core hosts three physics and three problem structures. Reviewers of that note correctly observed that it asserts a capability without exposing it: the residual equations, the boundary-condition treatment, and the closure models were not shown, and the VERA comparison displayed netflow output beside a qualitative statement of agreement rather than a quantitative error. This document supplies the missing methods and defines the quantitative validation. It is written so that an independent practitioner can reconstruct the formulation and judge, in numbers, how well it performs.

2 Mathematical formulation

2.1 Nodes, edges, and the nodal residual

The state of the system is a vector $\mathbf{x}$ of one scalar per *node*. An *edge* $e = (a, b)$ carries a flux $f_e(x_a, x_b)$ that depends only on its two endpoint states; the sign convention is that positive flux is transport $a \rightarrow b$. Conservation at every *interior* node i (one not pinned by a Dirichlet condition) is

$$F_i(\mathbf{x}) = s_i + \sum_{e=(a,b)} \left[\mathbb{K}_{b=i} f_e(x_a, x_b) - \mathbb{K}_{a=i} f_e(x_a, x_b) \right] = 0, \quad (1)$$

where s_i is a fixed nodal source (a Neumann term). Equation (1) is the only equation the core knows. All physics enters through the functional form of f_e and through s_i and the Dirichlet set; the core never inspects units or domain.

2.2 Solution by damped Newton

Let $J(\mathbf{x}) = \partial F / \partial \mathbf{x}$. For an edge (a, b) with analytic partials $(d_a, d_b) = (\partial f_e / \partial x_a, \partial f_e / \partial x_b)$, the four interior contributions are

$$\frac{\partial F_a}{\partial x_a} = -d_a, \quad \frac{\partial F_a}{\partial x_b} = -d_b, \quad \frac{\partial F_b}{\partial x_a} = +d_a, \quad \frac{\partial F_b}{\partial x_b} = +d_b, \quad (2)$$

with Dirichlet endpoints contributing only to F (their columns are absent from J). J is assembled as COO triplets, converted to CSC, and the Newton update $J \boldsymbol{\delta} = -F$ is solved with a sparse LU (`scipy.sparse.linalg.spsolve` [10]). A backtracking line search accepts the step $\mathbf{x} \leftarrow \mathbf{x} + \alpha \boldsymbol{\delta}$, halving α from 1 until $\|F\|_\infty$ decreases (or $\alpha \leq \alpha_{\min}$). Convergence is $\|F\|_\infty \leq \text{tol}$. A rank-deficient J is raised as a typed `SingularJacobian` rather than silently mis-solved.

2.3 Picard fallback

If *any* edge cannot supply an analytic Jacobian, the solve downgrades to Picard rather than filling zeros. Each edge is linearized as an effective conductance $g_e = f_e(x_a, x_b) / (x_a - x_b)$ (falling back to the local slope when $|x_a - x_b|$ underflows), assembling a weighted graph Laplacian L with Dirichlet states moved to the right-hand side; $L \mathbf{x}_{k+1} = \mathbf{b}$ is solved to a fixed point, optionally under-relaxed. Picard is slower on strong nonlinearity but always available, which keeps the core able to host an edge type before its derivative is implemented.

2.4 The outer-loop meta-abstraction

The core solves *driven* problems $F(\mathbf{x}) = 0$. Three larger problem classes are each an outer loop whose every step is a driven core solve:

- **Transient** $C_i \dot{x}_i = F_i$: implicit BDF time stepping [10]; each stage is a driven Newton solve. The analytic sparse Jacobian is essential—a dense $N \times N$ Jacobian costs $\mathcal{O}(N^2)$ memory (15 GB at a full assembly) and is the difference between a 4.2 s and a 320 s transient on a 7×7 array.
- **Eigenvalue** $M\varphi = k^{-1}P\varphi$: power iteration; each step is a driven diffusion solve with the fission source rebuilt from the previous flux (Section 3.3).
- **Coupled multiphysics**: Picard between single-physics solves, each a driven core solve.

All three reduce to one driver, `core.iterate.fixed_point`: $x_{n+1} = \text{step}(x_n)$ until $\|x_{n+1} - x_n\| < \text{tol}$. The power-iteration solver is built directly on it.

3 Governing residuals by domain

Each plugin supplies only edge flux functions f_e (and their partials). The network solver and equation (1) are unchanged across all three.

3.1 Thermal

State is temperature T (K); flux is heat rate (W).

$$\text{planar conduction: } f = k(\bar{T}) \frac{A}{L} (T_a - T_b), \quad (3)$$

$$\text{radial (cylinder): } f = \frac{2\pi L k(\bar{T})}{\ln(r_o/r_i)} (T_a - T_b), \quad (4)$$

$$\text{solid pellet (uniform } q'''): f = 4\pi k(\bar{T}) L (T_c - T_s), \quad (5)$$

$$\text{gray radiation: } f = \sigma \varepsilon_{\text{eff}} \mathcal{F} A (T_a^4 - T_b^4), \quad (6)$$

$$\text{upwind advection: } f_{a \rightarrow b} = \dot{m} c_p (T_a) T_a, \quad (7)$$

with $\bar{T} = \frac{1}{2}(T_a + T_b)$. Equation (5) is the finite-edge form of the closed solid-cylinder result $\Delta T_{c \rightarrow s} = q''' r^2 / 4k$ (derivation in Section 5). Substituting the upwind flux (7) into the residual (1) at an interior coolant slice k that receives wall heat $Q_{w,k}$ recovers the exact 1-D energy balance

$$Q_{w,k} + \dot{m} c_p (T_{k-1} - T_k) = 0 \implies T_k = T_{k-1} + \frac{Q_{w,k}}{\dot{m} c_p}, \quad (8)$$

i.e. the discrete enthalpy rise emerges from conservation, not a separate equation.

3.2 Hydraulic

State is pressure P (Pa); flux is volumetric flow Q (m³/s). The Darcy–Weisbach head loss $\Delta P = K Q |Q|$ inverts to

$$Q(P_a, P_b) = \text{sign}(\Delta P) \sqrt{|\Delta P|/K}, \quad \Delta P = P_a - P_b, \quad (9)$$

with $K = f \frac{L}{D} \frac{\rho}{2A^2}$ or supplied directly. The derivative $dQ/d\Delta P = 1/(2\sqrt{K|\Delta P|})$ is singular at $\Delta P = 0$ (a harder nonlinearity than radiation). We regularize with a linear laminar core of half-width ΔP_ϵ matched in value and slope at $\pm \Delta P_\epsilon$, keeping flux and Jacobian finite and continuous through zero—standard practice in pipe-network solvers.

3.3 Neutronic

State is scalar flux φ ; one-group diffusion. Leakage and removal are edges,

$$\text{diffusion: } f = \frac{DA}{\Delta} (\varphi_a - \varphi_b), \quad \text{absorption: } f = (\Sigma_a V) \varphi_{\text{cell}}, \quad (10)$$

the latter an edge from each cell to a Dirichlet $\varphi=0$ ground node (the network idiom for “loss proportional to state”). Criticality is the generalized eigenproblem $M\varphi = k^{-1}P\varphi$ solved by power iteration on the outer loop of Section 2.4:

1. fission source $S_i = k_n^{-1} \nu \Sigma_f V \varphi_{n,i}$ (a Neumann source);
2. driven core solve $M\varphi_{n+1} = S$;
3. eigenvalue update $k_{n+1} = k_n \langle P\varphi_{n+1} \rangle / \langle P\varphi_n \rangle$;
4. normalize φ and repeat.

The continuous bare-slab reference is $k = \nu \Sigma_f / (\Sigma_a + DB^2)$, $B = \pi/L$.

4 Boundary conditions

Boundary conditions are imposed through node and edge structure, not special-cased in the solver:

Dirichlet a node flagged **fixed** holds a prescribed state; it is removed from the unknown vector and enters neighbors' residuals as a constant. Used for fixed temperatures, inlet/outlet plenum pressures, and the $\varphi=0$ vacuum/ground boundaries.

Neumann a nodal source s_i in (1): imposed heat (linear power deposited at a fuel centerline), or the fission source in power iteration.

Advective outflow the last coolant slice connects to a Dirichlet sink via the upwind edge (7); because that flux depends only on the upstream state, the sink absorbs the correct outflow regardless of its own value—a natural outflow condition with no artificial gradient.

Vacuum (neutron) cell-center-to-face diffusion edges to $\varphi=0$ boundary nodes, using a half-cell spacing $\Delta/2$, giving the zero-flux extrapolated boundary.

Eigenvalue closure criticality has no boundary state; the eigenvalue k is carried in the outer-loop closure and updated from the fission-source ratio, leaving each inner solve a standard driven problem.

5 Closure models

Solid conductivity $k(T)$. Conduction edges evaluate k at the edge mean temperature $\bar{T}$ and carry dk/dT in the Jacobian. UO_2 uses the Fink fit [4],

$$k_{\text{UO}_2}(T) = \frac{1}{0.0375 + 2.165 \times 10^{-4}T} + \frac{4.715 \times 10^9}{T^2} e^{-16361/T} \text{ W/m/K}, \quad (11)$$

(phonon term plus high-temperature electronic term); Zircaloy-4 uses $k = 12.6 + 0.0048(T - 300)$ [5].

Limitation: the two-node pellet evaluates k at a single mean temperature rather than integrating the conductivity integral $\int k(T) dT$ across the radius; since k_{UO_2} falls $\sim 3\times$ between 500 K and 2000 K, this linearizes the profile and under-resolves centerline at high local power. Radial meshing is a fidelity dial within the same abstraction (more nodes/edges along r), not an architectural change.

Fuel pellet. For a solid cylinder of radius r with uniform q''' , $\Delta T_{c \rightarrow s} = q'''r^2/4k$. With linear power $q' = \pi r^2 q'''$ this is $R_{\text{pellet}} = 1/(4\pi kL)$, giving the edge flux (5).

Fuel-clad gap. Two interchangeable models. (i) *Geometric*: helium gas conduction across the annulus ($k_{\text{He}} = 2.682 \times 10^{-3} T^{0.71}$ [6]) in parallel with gray-body radiation. (ii) *Empirical*: a gap conductance h_{gap} as a contact resistance $R = 1/(h_{\text{gap}} A_{\text{gap}})$, $A_{\text{gap}} = 2\pi r_p L$. A controlled comparison showed the geometric model over-predicts fuel centerline by ~ 220 K (an open hot gap with no contact or temperature-jump terms); the VERA runs use the empirical form, $h_{\text{gap}} = 7500 \text{ W/m}^2/\text{K}$. This is a fixed input, not a mechanistic gap model (no burnup, gap closure, or fission-gas effects) [9].

Single-phase convection. Clad-to-coolant flux is $h A_{\text{ht}} (T_w - T_b)$ with $h = \text{Nu } k_f / D_h$. Dittus-Boelter [7], $\text{Nu} = 0.023 \text{Re}^{0.8} \text{Pr}^n$ ($n = 0.4$ heating, 0.3 cooling), or Gnielinski [8] for wider Re range; $\text{Re} = \dot{m} D_h / (\mu A_{\text{xs}})$, properties at film temperature from CoolProp [11]. h is frozen within a Newton step and recomputed between steps. *Limitation:* single-phase liquid only—no subcooled nucleate boiling, no DNB/CHF.

Table 1: Neutron-eigenvalue spatial convergence to the analytic bare-slab k_∞ . Error is second order in cell count.

cells N	error (pcm)	ratio
20	9.7	—
50	1.5	$6.5\times$
100	0.4	$3.8\times$
200	0.1	$4.0\times$

Lateral coolant transport and cross-flow. Two mechanisms. (i) *Turbulent enthalpy mixing*: a symmetric edge $f = \dot{m}_{\text{mix}} c_p (T_a - T_b)$ between adjacent channels, $\dot{m}_{\text{mix}}$ a calibrated fraction of axial flow—no lateral mass transfer. (ii) *Pressure-driven cross-flow*: in the subchannel model each channel has axial Darcy segments (9) plus lateral Darcy edges to its four neighbors; solving the 3-D pressure field redistributes mass (low-resistance guide tubes draw bypass flow, starving neighbors), Picard-coupled to the thermal solve through density $\rho(T) = 745 - 1.55(T - 565) \text{ kg/m}^3$. *Limitation*: this is a scalar pressure-resistance network, *not* the transverse momentum equation—no lateral inertia, momentum flux, turbulent momentum mixing, void drift, or grid-spacer cross-flow. The coefficients (axial/lateral K , mixing fraction) are calibrated, as in production subchannel codes [2].

6 Verification

Verification asks whether the equations are solved correctly (independent of whether they describe a reactor). Evidence:

- **Edge resistances** reproduce their analytic closed forms to machine precision; the fuel-pin chain matches the textbook radial decomposition (film, gap, clad, fuel) [9].
- **Analytic Jacobians** agree with finite differences to $\leq 10^{-5}$ (relative), edge by edge.
- **Neutron eigenvalue** converges to the analytic bare-slab value at second order in mesh spacing (Table 1); the fundamental mode matches the analytic cosine.
- **Global energy balance** on a full assembly closes to six significant figures (a global invariant, not a per-edge check—verification vs. validation).
- **Automated suite**: 107 tests across core, the three plugins, the coupling layer, and a CI “layering” test enforcing that the core has zero domain imports.

7 Code-to-code comparison against VERA

What VERA P6/P7 are. The VERA specification states verbatim, for both Problem 6 and Problem 7, “No reference solution exists for this problem at this time” [1, pp. 79, 81]: these problems define *inputs only*. Every published P6/P7 *result* is a code solution (MC21/CTF, MC21/COBRA-IE, VERA/MPACT [2]), not a measured or reference truth. We therefore frame this section honestly as a *code-to-code comparison* against those published solutions, not validation against a benchmark standard. (Reference-grade KENO-VI eigenvalues exist only for the simpler Problems 1–5.)

Table 2: Code-to-code comparison against published VERA P6/P7 solutions. VERA P6/P7 have *no* reference solution; the third column is another code’s result, cited. Reproducible via `netflow.bench.vera.codecompare`.

quantity (definition)	netflow	published [2]	note
peak <i>volume-avg</i> fuel temp. ($^{\circ}\text{C}$)	999	1066	-6.2%
local 3-D pin peaking	1.918	1.92	input
coolant exit spread (K)	4.9–6.5	6.6 / 8.7	calibrated
coolant mean rise (K)	37–39	34–37	—
eigenvalue k	analytic, Table 1	—	verification only
fuel- T Doppler (pcm/K)	negative (sign)	-2 to -2.5	mechanism

netflow vs published VERA P6/P7 code solutions (code-to-code; P6/P7 have no reference solution)

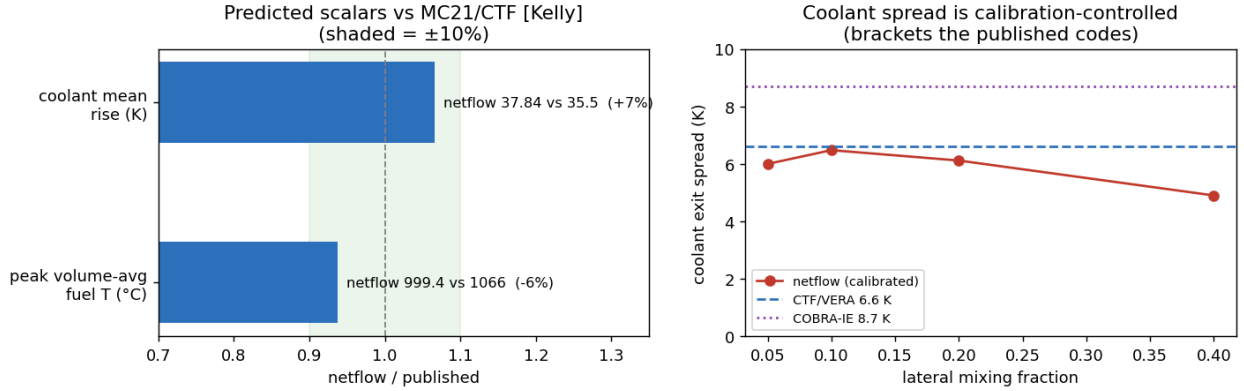


Figure 1: Code-to-code comparison, visualized. *Left:* predicted scalars as the netflow/published ratio (shaded $\pm 10\%$) — peak volume-average fuel temperature and coolant mean rise both within $\sim 7\%$ of the MC21/CTF solution [2]. *Right:* the calibrated coolant exit spread vs. the lateral-mixing fraction, bracketing the published CTF/VERA (6.6 K) and COBRA-IE (8.7 K) [3] values — the spread is calibration-controlled, not predicted.

Statepoint. Problem 6, from the specification [1]: Westinghouse 17×17 (264 fuel rods, 24 guide + 1 instrument tube), pellet OD 0.8192 cm, clad OD 0.95 cm, gap 84 μm , rod pitch 1.26 cm, active length 365.76 cm; inlet 565 K, 15.51 MPa, 1300 ppm boron, 3.10 wt% enrichment; rated assembly power 17.67 MW (97.4% in fuel); assembly flow 85.98 kg/s. These give an average linear heat rate of 178 W/cm, matching the spec.

Quantities at their published definition. Two definition corrections matter, and were the source of the note’s overstated claims. VERA reports *volume-average* fuel temperature (not centerline; the parabolic-pellet volume average is $(T_c + T_s)/2$), and the coolant “spread” is read off the published subchannel field [2, Fig. 11], not a headline scalar. Table 2 compares each quantity at its published definition; “input” marks a quantity prescribed to netflow rather than predicted, “calibrated” a quantity set by a tuned coefficient.

Reading the table honestly. Fuel temperature is the strongest comparison (Fig. 1): netflow’s peak volume-average (999 $^{\circ}\text{C}$) lands within 6% of the published MC21/CTF value (1066 $^{\circ}\text{C}$) for a

single hot pin at Problem 7’s peak local peaking (1.918 vs. the published 1.92), given the calibrated gap conductance. The coolant mean rise matches the assembly energy balance, $17.21 \text{ MW}/(\dot{m}c_p) \approx 36 \text{ K}$. The exit spread is *not* a prediction—it is set by the cross-flow calibration—but for the Problem 6 (single enrichment, near-flat power) assembly, sweeping the lateral mixing fraction over 0.05–0.40 gives 4.9 K to 6.5 K, essentially matching the MC21/CTF and VERA value (6.6 K) and falling under the MC21/COBRA-IE value (8.7 K) [3]. The eigenvalue row is *verification* against an analytic slab, not a P6 comparison: netflow’s one-group diffusion does not attempt the multigroup pin-resolved P6 eigenvalue (1.16424 [2]), and we do not claim it.

Self-consistent multiphysics (mechanism). Coupling the neutronic and thermal plugins through Picard with Doppler feedback ($\Sigma_a \propto \sqrt{T}$) *computes* the power distribution rather than imposing it, closing the “power is imposed” caveat. The coupled solve exhibits a *negative* fuel-temperature reactivity coefficient—the sign underlying inherent reactor stability. We report this as a mechanism/sign demonstration only: it is shown on a one-group bare slab with illustrative cross-sections, so its *magnitude* is not a physical prediction. The physical reference for fresh 3.10 wt% UO_2 , -2 to -2.5 pcm/K , is itself *derived* by us from the spec’s reference KENO-VI eigenvalues at varied fuel temperature for Problems 1–2 [1] (the spec reports no Doppler coefficient directly).

8 Scope and limitations

Relationship to established acausal tools. The abstraction here is not new. A node carrying a *potential* (across) variable and an edge carrying a conserved *flow* (through) variable, with potentials equated and flows summed to zero at each node, is the acausal connector formalized by Modelica [12] and, in Julia, ModelingToolkit [13] (productized as Dyad), on the older effort/flow foundation of bond graphs [14]. Those ecosystems are mature, symbolic, compiled, and far faster, with validated domain libraries (e.g. TRANSFORM for nuclear thermal-hydraulics). **netflow** is best understood as a *minimal, transparent, pure-Python re-derivation* of that paradigm—useful for pedagogy and dependency-light embedding, not a new modeling capability nor a replacement for those tools.

The defining boundary: **netflow** is a scalar conservation-network solver. Every edge flux is a function of two scalar endpoint states, and there is no momentum equation—pressure drop is a quasi-steady loss relation, not solved momentum. Consequences and honest growth paths:

item	status	cost in this abstraction
conductivity-integral pellet	two-node mean- T now	fidelity dial (radial mesh)
two-phase convection / DNB	single-phase only	fidelity dial (new edge correlations)
transverse momentum closure	scalar pressure network	outside the scalar-flux abstraction
mechanistic gap (burnup)	fixed h_{gap}	external correlation input

The first two and the gap are fidelity dials inside the current model; transverse momentum is the one genuine frontier. **netflow** targets fast, transparent pre-design scoping accurate to tens of degrees, on a single core that also solves pipe networks and criticality eigenproblems. It is not a replacement for CTF/COBRA-TF.

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